MIL-A-18977D(OS)
7 June 1971

Superseding
MIL-A-18977C(OS)
7 April 1967

MILITARY SPECIFICATION

AMPLIFIER CHANNEL

(For Torpedo Mark 37 Mods 0, 2, and 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement, inspection and test of an Amplifier Channel, LD 162565, LD 296726, and Drawing 1154877.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

NIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40 " x 48 " Pallets, Skids, Runners, or Pallet-Type Base

FSC 1355

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DRAWINGS

Naval Ordnance Systems Command

LD 162565	Servo Amplifier Assembled, and all parts and assemblies listed thereon (Mods 2 and 3)
LD 296726	Servo Amplifier Assembled, and all parts and subassemblies listed thereon (Mod 0)
LD 162754	Shipping (Packing) Condition, For One Servo Amplifier Assembly
NAVORD Dwg 694739	Actuator Assembled
NAVORD Dwg 696886	Servo Amplifier Assembled (Mods 2 and 3)
NVORD Dwg 859987	Board #2 Subassembly
NAVORD Dwg 877407	Servo Amplifier Assembled (Mod 0)
NAVORD Dwg 877408	Schematic Wiring Diagram
NAVORD Dwg 1154877	Amplifier Channel Subassembly

PUBLICATIONS

Naval Ordnance Systems Command

NAVORD OS 6341	Miscellaneous General Ordnance Design Requirements

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a paart of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal Shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for Federal Services

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Description. The amplifier channel is an electronic amplifier which receives a 400-Hertz error voltage and amplifies this error voltage in a phase-sensitive detector, the dc output of which is fed to a dc voltage amplifier and to a power amplifier. The output of the power amplifier controls an electromagnetic clutch, Drawing 694739.

3.2 General requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the amplifier channel shall be manufactured using methods and procedures proposed for the production lot. The units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractors production methods are capable of producing amplifier channels that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the amplifier channel shall conform to the detailed requirements indicated on the drawings and specifications listed in LD 162565, LD 296726, and Drawing 1.154877.

3.2.4 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for the amplifier channel shall be in accordance with the applicable drawings, specifications, and publications listed in LD 162565, LD 296726, and Drawing 1154877.

3.2.5 Moisture, fungus, and corrosion. The amplifier channel shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified in the drawings and specifications listed in LD 162565, LD 296726, and Drawing 1154877. When moisture, fungus, and corrosion resistance are not specified in these documents, the amplifier channel shall conform as required by NAVORD OS 6341.

3.2.6 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in LD 162565, LD 296726, and Drawing 1154877.

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3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for-threads shall be as specified on the applicable drawings,

3.2.8 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Performance requirements and product characteristics. The amplifier channels shall meet the following performance requirements and product characteristics. Unless otherwise specified, all terminals referred to hereafter are listed on Drawing 877408. All three amplifier channels shown on Drawing 877408 are identical. All references to terminal connections in this test refer to connector P401, but they also apply equally to connectors P402 and P403.

3.3.1 Loads. A resistive load of 5700 ohms (4.5.3,3.2) shall be connected from each output terminal (R and S) to terminal L.

3.3.2 Phase relationships. When a 400-Hertz (Hz) signal is applied to terminals H and K so that the polarity at terminal H corresponds to the polarity at terminal A, the output current from terminal R shall increase and the current from terminal S droll decrease.

3.3.3 Balance currents. With the output currents from terminals R and S balanced, the balance current per plate shall be 6.5 plus or minus 1.5 milliamperes (ma).

3.3.4 Amplifier gain. With the output currents balanced, a 0.1 volt signal of the same phase as the 400 Hz power source, applied to terminals H and K, shall result in a current increase from balance to 13 plus 1 ma minus 1.5 ma from one output terminal (R or S), and a current decrease from balance to 3 ma maximum from the other output terminal.

3.3.5 DC signal gain. With the output currents balanced a direct current (de) voltage of 1.5 volts applied at terminals M (+) and N (-) shall result in an output current change from balance to 3.5 ma maximum from terminal R, and from balance to 10.5 ma minimum from terminal S.

3.3.6 Current drain. The dc current drawn from the 150-volt power source (4.5.3.1.1) shall not exceed 18.5 ma.

3.3.7 Operating life. The amplifier channel, after testing, shall be capable of at least 200 hours of operation without servicing.

3.4 Environmental requirements. The amplifier channel shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.4.

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3.4.1 Vibration. The amplifier channel shall meet the following vibration requirements:

3.4.1.1 Vibration, nonoperating. The amplifier channel shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.1.2.

3.4.1.2 Vibration, operating. The amplifier channel shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.4.1.3.

3.4.2 Temperature and humidity

3.4.2.1 High temperature and humidity, nonoperating. The amplifier channel shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.1.

3.4.2.2 High temperature, operating. The amplifier channel shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.2.2.

3.4.2.3 Low temperature, nonoperating. The amplifier channel shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.3.

3.4.2.4 Low temperature, operating. The amplifier channel shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.2.4.

3.5 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.6 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, impregnation of coils, plating, painting, machine screw assemblage, welding, riveting, brazing, and freedom of parts from all burrs and sharp edges.

The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as those standards are not specifically covered by the applicable specifications.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the amplifier channel shall be classified as follows:

(a) Preproduction inspection (paragraph 4.4)

(b) Quality conformance inspection (paragraph 4.5)

(c) Periodic production inspection (paragraph 4.4)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five amplifier channels, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the amplifier channel by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.5, a random sample of five amplifier channels will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The amplifier channels shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

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4.4 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded. to the procuring activity.

4.4.1 Vibration

4.4.1.1 Mounting. The amplifier channel, less pack or packaging, shall be mounted in the servo amplifier case. The case, mounted securely to a vibration device by means of its mounting holes, shall be tested as specified in 4.4.1.2 and 4.4.1.3.

4.4.1.2 Vibration, nonoperating. The amplifier channel shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.1.1.

4.4.1.3 Vibration, operating. The amplifier channel shall be subjected to the "Vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.4.1.2.

4.4.2 Temperature and humidty

4.4.2.1 High temperature and humidity, nonoperating. The amplifier channel shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.2.1.

4.4.2.2 High temperature, operating. The amplifier channel shall be subjected to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.4.2.2.

4.4.2.3 Low temperature, nonoperating. The amplifier channel shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.2.3.

4.4.2.4 Low temperature, operating. The amplifier channel shall be subjected to the "Low temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.4.2.4.

4.4.3 Operating life. To test for operating life as specified in 3.3.7:

4.4.3.1 With the amplifier at an ambient temperature of 65 degrees F to 95 degrees F (18 degrees C to 35 degrees C), and test preparations made as specified in 4.5.4.1, allow the amplifier channel to operate for a period of 30 minutes.

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4.4.3.2 Balance the amplifier channel as specified in 4.5.4.2.2, and allow to operate an additional 30 minutes, after which the power sources (4.5.3.1) will. be deenergized. Do not disturb the balancing adjustment for the remainder of the life test.

4.4.3.3 Cycle the amplifier for a period of one hour on and one hour off, for a total operating time of 200 hours (200 complete cycles).

4.4.3.4 At the conclusion of this test, the amplifier shall meet the performance requirements specified in 3.3.

4.5 Quality conformance inspection. Each amplifier channel delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirements specified herein shall be considered cause for rejection.

4.5.1 Test conditions. Unless otherwise specified, the amplifier channel shall be subjected to quality conformance tests under the following conditions:

4.5.1.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18 degrees C to 35 degrees C).

4.5.1.2 Altitude: Normal ground atmospheric pressure.

4.5.1.3 Vibration: None.

4.5.1.4 Humidity: Room ambient to 95 percent relative maximum.

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide all personnel for performing the quality conformance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests.

4.5.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.5.3.1 Power sources

4.5.3.1.1 One 150-volt dc plus or minus 3 percent, 20 ma, 0.05 volt ripple maximum.

4.5.3.1.2 One 6.3-volt ac plus or minus 5 percent, center tapped, 1.5 amperes, 400 Hz plus or minus 2 percent.

4.5.3.1.3 Two 100-volt ac plus or minus 3 percent, 400 Hz plus or minus 2 percent, 5 ma, isolated supplies.

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4.5.3.1.4 One variable 3-volt dc plus or minus 10 percent, 10 ma.

4.5.3.1.5 One 400 Hz, phase reversible, continuously variable, signal source of the same phase as the 400 Hz power sources (4.5.3.1.3) and capable of voltage changes of 0.01 volt and a maximum of 5 volts.

4.5.3.2 Measuring equipment

4.5.3.2.1 Two milliammeters, capable of indicating current values from 0 to 15 ma to an accuracy of 0.5 ma.

4.5.3.2.2 One milliammeter, capable of indicating current values from 0 to 20 ma.

4.5.3.2.3 One ac vacuum tube voltmeter with the following scales: 0 to 0.1, 0.3, 1.0 volts.

4.5.3.3 Miscellaneous components

4.5.3.3.1 One female connector (S401 on Drawing 859987).

4.5.3.3.2 Two 5700 ohm plus or minus 1 percent resistors, rated at least 5 watts.

4.5.4 Test procedure

> CAUTION: In performing the tests specified in 4.5, extreme caution should be exercised, as dangerous voltages are present which may cause bodily injury or death.

4.5.4.1 Test preparations. The following test preparations are required to perform the quality conformance tests specified herein. Unless otherwise specified, all terminals designated herein are listed on Drawing 877408. All three amplifier channels are identical, and all references to terminals on P401 hold equally for P402 and P403, as in 3.3

4.5.4.1.1 Connect the positive terminal of a milliammeter (4.5.3.2.1) to a 5700 ohm resistor (4.5.3.3.2).

4.5.4.1.2 Connect the negative terminal of the milliammeter to terminal R.

4.5.4.1.3 Connect the other end of the resistance to terminal L.

4.5.4.1.4 Connect the positive terminal. of the second milliammeter (4.5.3.2.1) to the second 5700 ohm resistance (4.5.3.3.2).

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4.5.4.1.5 Connect the negative terminal of the second milliammeter to terminal S.

4.5.4.1.6 Connect the other end of the second resistance to terminal L.

4.5.4.1.7 Apply the 6,3 ac power source (4.5.3.1.2) to terminals E and F. Center tap to terminal P.

4.5.4.1.8 Apply one of the 100-volt ac power sources (4.5.3.1.3) to terminals B and C.

4.5.4.1.9 Apply the other 100-volt ac power source (4.4.3.1.3) to terminals A and D.

NOTE: Terminals A and B shall have corresponding polarities, as shall terminals C and D.

4.5.4.1.10 Apply power (4.5.3.1.1) to terminals L and P (L being positive).

4.5.4.2 Balance currents. To test for balance currents as specified in 3.3.3:

4.5.4.2.1 Apply a signal source (4.5.3.1.5) to terminals H and K.

4.5.4.2.2 Adjust this signal, both in magnitude and polarity, until the output currents from pins R and S are balanced at 6.5 plus or minus 1.5 ma.

4.5.4.3 Amplifier gain. To test for amplifier gain as specified in 3.3.4:

4.5.4.3.1 Adjust the magnitude of the signal voltage (4.5.3.1.5) for a change of 0.1 volt from the balanced condition (4.5.4.2.2).

4.5.4.3.2 The output current from one terminal (R or S) shall show an increase to 13 plus 1 ma minus 1.5 ma, and the output current from the other terminal (R or S) shall show a decrease to 3 ma maximum, as specified in 3.3.4.

4.5.4.3.3 Adjust the signal source (4.5.3.1.5) to give a 1 plus 0.1 volt signal with polarity at terminal H the same as at terminals A and B; and the polarity of terminal K the same as at terminals C and D, as per 3.3.2. The output current from terminal R shall read 13 plus 1 ma minus 1.5 ma, and the output current from terminal S shall read 3 ma maximum.

4.5.4.3.4 Readjust the signal as specified in 4.5.4.2.2.

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4.5.4.4 DC signal gain. To test for dc signal gain as specified in 3.3.5:

4.5.4.4.1 Apply power signal source (4.5.3.1.4) to terminals M and N (M being positive).

4.5.4.4.2 Adjust the magnitude of the power signal source to 1.5 volts.

4.5.4.4.3 The resulting current shall show an increase from balance to 10.5 minimum, from terminal S, as specified in 3.3.5.

4.5.4.4.4 The resulting current shall show a decrease from balance to 3.5 ma maximum, from terminal R, as specified in 3.3.5.

4.5.4.4.5 Remove power (signal) source (4.5.3.1.4).

4.5.4.5 Current drain. To test for current drain as specified in 3.3.6:

4.5.4.5.1 Remove the power supply lead from terminal L.

4.5.4.5.2 Connect the lead removed from terminal L to the positive terminal of a 0-20 ma dc milliammeter (4.5.3.2.2).

4.5.4.5.3 Connect a lead from the negative terminal of the meter to terminal L.

4.5.4.5.4 With the input from the signal source (4.5.3.1.5) at any polarity and magnitude, the current indicated by this meter shall not exceed 18.5 ma, as specified in 3.3.6.

4.5.4.6 Visual and dimensional examination. Each amplifier channel shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.5.4.7 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2)

5.1.1 Level A

5.1.1.1 Unit packaging. The amplifier channel shall be packaged in accordance with Method IID of MIL-P-116 and LD 162754.

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5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The amplifier channel shall be packaged in accordance with Method III of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Number of packages per pack. The number of unit packages per pack, for Level C, shall be as specified in the contract or order.

5.2.2 Level A

5.2.2.1 Unit pack. The amplifier channel shall be packed in accordance with LD 162754 and may be shipped individually in small lots.

5.2.2.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Load Type III (see 6.2).

5.2.3 Level B. Not applicable.

5.2.4 Level C. The amplifier channel shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Normal markings, Level A. Unit packs shall be marked in accordante with LD 162754.

5.3.2 Normal markings, Level C. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.3 Serial numbers. Unit packages and shipping containers shall be marked with the serial numbers of the units contained within.

6. NOTES

6.1 Intended use. The amplifier channel is intended for use in Torpedo Mk 37 Mods 0, 2, and 3.

6.2 Ordering data. Procedurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

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(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) Data required other than that specified in 3.2.1.

(g) That addition samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

Preparing Activity
Navy-OS
Project No. 1355-N178

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